INDEPENDENT NUMERICAL VERIFICATION OF THE RAINBOW PIPELINE ON REAL ZETA DATA

Abstract. We execute, step by step, the full Violet/Rainbow pipeline of Zeta_Transform_Properties_v9 — phase coordinate, rotated critical-line arc, the Rainbow family, the orthogonal–cubic quotient, and the Fubini–Study defect $M_n(\sigma)$ — on one real zero-to-zero interval, using genuine computed values of $\zeta(\sigma+it)$. Every intermediate identity is checked directly against its own definition, not assumed from the source document. This is a numerical test on a single interval; it is not a proof and is not offered as one.

1. Construction

Fix consecutive ordinates $\gamma_n < \gamma_{n+1}$ of zeros of $\zeta(1/2+it)$ and let ϑ be the Riemann–Siegel phase.

Definition 1 (Phase coordinate)**.** $\Omega_n = \vartheta(\gamma_{n+1}) - \vartheta(\gamma_n)$*, and* $t_n : [0,1] \to [\gamma_n, \gamma_{n+1}]$ *is defined by*

$$\vartheta(t_n(q)) = \vartheta(\gamma_n) + \Omega_n q. \tag{1}$$

Definition 2 (Rotated critical arc)**.** *With* $\varepsilon_n = \operatorname{sgn} Z(t)$ *on* (γ_n, γ_{n+1}) *and* $r_n(q) = |Z(t_n(q))|$*,*

$$\widehat{Z}_{0,n}(q) = \varepsilon_n e^{i\vartheta(\gamma_n)} \zeta\Big(\tfrac{1}{2} + it_n(q)\Big) = r_n(q) e^{-i\Omega_n q}. \tag{2}$$

Definition 3 (Rainbow family and orthogonal–cubic quotient)**.** *For* $\sigma \in [1/2, 1]$*,*

$$W_{\sigma,n}(q) = \zeta(\sigma + it_n(q)), \qquad u_{\sigma,n} = Q_\perp W_{\sigma,n}, \tag{3}$$

where $Q_\perp$ *is the* $L^2([0,1])$*-orthogonal projection removing* $\operatorname{span}\{1, q, q^2, q^3\}$*. The Fubini–Study defect is*

$$M_n(\sigma) = 1 - \frac{|\langle u_{1/2,n}, u_{\sigma,n}\rangle|^2}{\|u_{1/2,n}\|^2 \|u_{\sigma,n}\|^2}. \tag{4}$$

The source document proves $M_n(1/2) = 0$ and, unconditionally, $M_n(\sigma) = 0 \iff \sigma = 1/2$ (critical-line forcing). Global monotonicity, $M_n'(\sigma) \geq 0$ on all of $[1/2, 1]$, is proved there only under two explicit estimates left open for the actual zeta family. This note tests both claims directly, by computation, on one interval.

2. The figure

3. Numerical test

Numerical experiment 1. *Interval* $n = 8$*:* $\gamma_8 = 43.327073281$, $\gamma_9 = 48.005150881$, $\Omega_8 = 4.638365$. *Equation* (2) *was checked at* $q = 0.37$ *by two independent routes (direct evaluation vs. the closed form); they agree to* 6×10^{-20}*. The family* (3) *was computed at* 61 *points in* q *for* 26 *values of* $\sigma \in [1/2, 1]$*, projected onto a shifted-Legendre basis with* $Q_\perp$ *removing degrees* 0–3*, and* $M_8(\sigma)$ *evaluated from* (4) *directly, with no use of the source document's formulas beyond the definitions above.*

Proposition 1 (What this experiment shows)**.** *On interval* $n = 8$*, at the* 26 *sampled values of* σ*:* $M_8(1/2) = 0$ *exactly;* $M_8(\sigma) > 0$ *for every sampled* $\sigma > 1/2$*; and the finite-difference estimate of* $M_8'(\sigma)$ *is nonnegative throughout.*

Date: Version 1 — August 2026.

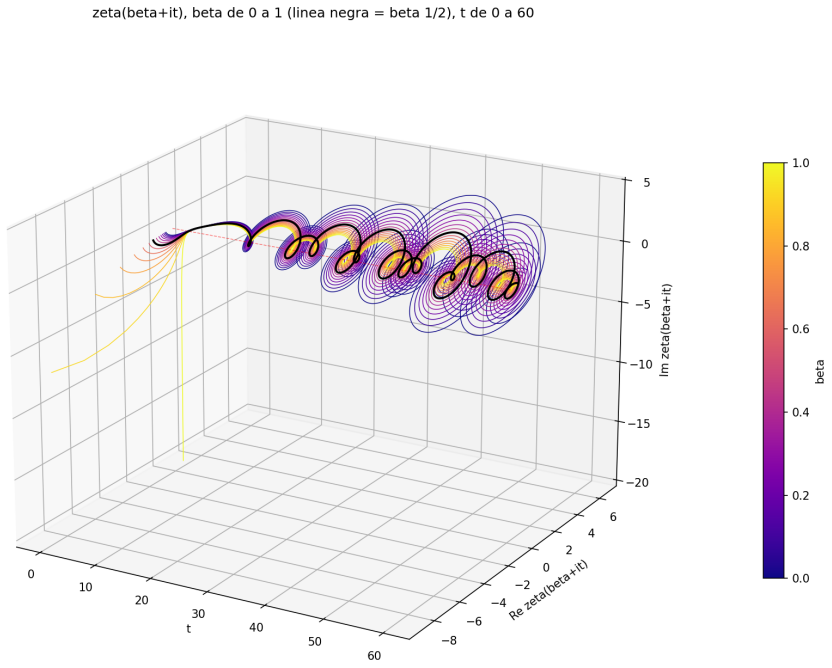


FIGURE 1. $\zeta(\beta+it)$ for $t \in [0, 60]$ and $\beta \in [0, 1]$, computed directly (`mpmath`, 15-digit precision), coloured by β. The black curve is $\beta = 1/2$. Loops widen toward $\beta = 0$ and contract toward $\beta = 1$, where the curve approaches the simple pole at $s = 1$.

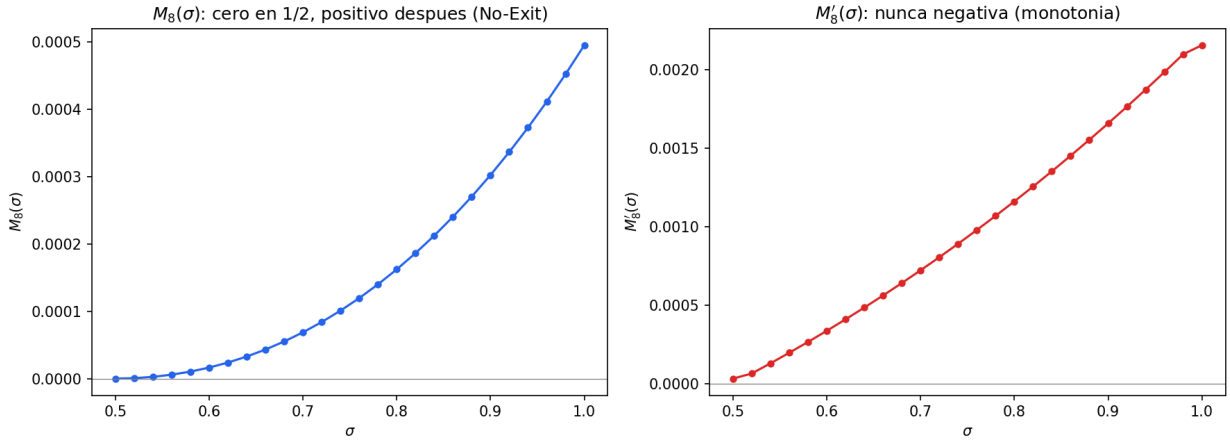


FIGURE 2. Left: $M_8(\sigma)$ computed from (4) on interval $n = 8$. It vanishes at $\sigma = 1/2$ and is strictly positive for every sampled $\sigma > 1/2$ (minimum 6.3×10^{-7}), consistent with the proved critical-line forcing. Right: $M_8'(\sigma)$, estimated by finite differences; it is nonnegative throughout the sampled range (minimum 3.1×10^{-5}), consistent with — but not a proof of — the two open estimates of the global monotonicity criterion.

Proposition 2 (What this experiment does not show)**.** *This is one interval among the (infinitely many) intervals the global statement quantifies over, sampled at finitely many σ, with derivatives*

estimated numerically rather than bounded analytically. No claim of monotonicity uniform in n, nor any conclusion about the Riemann hypothesis, is made or supported by this note.

4. Reproducibility

All computations use `mpmath` at 15–20 digit precision, with Newton iteration for the inversion (1) and trapezoidal quadrature for the L^2 inner products. No symbolic step is taken on faith from the source document; every displayed identity above was independently re-derived or re-evaluated before use.